

AFCI-Bench - results delivery

Compiled 2026-09-18 from study-results/ on branch study-v2 at 0dc81a0b3d6d.

Reporting and export only. No benchmark observation was executed to produce this package, no model was invoked, and no task definition, architecture document, scorer, threshold, run plan, condition, raw run artifact or prior analysis was changed. Every figure below was recomputed from the 121 run/attempt rows and then checked against the frozen per-experiment analysis artifacts:

124 checks, 0 mismatches. The full check list is in the workbook sheet 20_RECOMPUTATION_AUDIT.

Claims are tagged **[FACT]** (a measured value, reproducible from the artifacts),

[INTERPRETATION] (a reading of those facts) and **[LIMITATION]** (a bound on what the facts can support).

1. Study objective

AFCI asks a narrow question. If you hand a coding agent an explicit, machine-readable description of a repository's architecture - the MAD - does it build better, or more cheaply, than an agent given the task alone?

The programme operationalises that as two conditions held identical in every other respect:

- **C1** - the task, and nothing else.
- **C4** - the same task, plus the explicit Machine-readable Architecture Document.

Two constructs are measured, and they are never combined into one score:

- **architectural conformance** - does the run violate the architecture

opportunities its own changes made relevant? This is the study's actual construct.

- **efficiency** - tokens, wall time, exploration, tool effort and provider cost.

This is the secondary construct, and it is the one with a clean pilot.

2. Experiments conducted

#	experiment	model	runs	status	decision
1	V1 original study	"Opus 7"	48	COMPLETE, PUBLISHED, superseded	published, then superseded by v1 limitations L1-L7
2	PT08 C1 difficulty diagnostic	claude-sonnet-5	3	COMPLETE	PT08 = REVISE; benchmark investment = CONTINUE
3	PT09 instrument qualification	claude-sonnet-5	3 (+1 infra-invalid)	COMPLETE	FAIL / ARCHITECTURE FLOOR -> STOP / REASSESS
4	PT10 instrument qualification	claude-sonnet-5	3	COMPLETE	REVISE / WEAK PRESSURE -> STOP / REASSESS
5	Efficiency pilot, Attempt 1	claude-sonnet-5	9 attempted of 36	ABORTED	excluded wholesale; no analysis was ever performed
6	Efficiency pilot, Attempt 2	claude-sonnet-5	36	COMPLETE	STOP - NO EFFICIENCY SIGNAL JUSTIFIES FULL-SUITE EXPANSION
7	Lower-capability model pilot	claude-haiku-4-5-2025 1001	18	COMPLETE	NO LOWER-MODEL SIGNAL - DO NOT EXPAND THE SYNTHETIC LOWER-MODEL MATRIX
8	Open-source complexity study	TBD	0	NOT STARTED	none - no runs exist and no result is pre-populated

Both completed pilots were judged by a decision rule frozen **before** the data they judge existed, and both rules returned a negative verdict. Neither was changed afterwards.

3. Evidence and run inventory

[FACT] Mechanically counted from AFCI_MASTER_RUN_RESULTS.csv.

measure	count
total run/attempt records	121
distinct run identities	118
completed observations	110
functionally valid observations	60
explicitly functionally invalid	2
no functional verdict captured at all	59
diagnostic / qualification observations	10
infrastructure-invalid or damaged records	3
refused	1
excluded from every analysis	71
eligible for any analysis	50
confirmatory observations	0

The three "missing" run identities are not a bookkeeping error: the PT08 diagnostic minted one run id for all three of its repetitions, and one Attempt-1 collision pair shares a single id. Both are recorded defects, kept as found.

experiment	rows	status COMPLETE	functionally valid	eligible	token coverage	cost coverage	architecture coverage
V1_ORIGINAL	48	48	not captured	0	0/48	0/48	0/48
V2_PT08_DIAGNOSTIC	3	3	3	0	0/3	0/3	3/3
V2_PT09_QUALIFICATION	4	3	3	0	0/4	0/4	3/4
V2_PT10_QUALIFICATION	3	3	3	0	0/3	0/3	3/3
V2_EFF_ATTEMPT1	9	0	not captured	0	0/9	0/9	0/9
V2_EFF_ATTEMPT2	36	35	34	34	35/36	19/36	0/36
V2_LOWER_MODEL_PILOT	18	18	17	16	18/18	18/18	18/18

V2_EFF_ATTEMPT1 shows 0 under *status COMPLETE* because its rows carry *INTACT_GOVERNED_OBSERVATION* (7 rows) and *DAMAGED_** (2 rows) instead - a deliberately distinct status, because those runs were never admitted as observations. The experiment registry counts the same 7 rows as "completed runs"; both readings are shown rather than reconciled away.

[LIMITATION] All 50 eligible rows belong to two

non-confirmatory pilots. No p-value, confidence interval, effect size or power estimate exists anywhere in this programme, and none would be defensible from 16 and 8 paired blocks at 3 repetitions.

4. V1 historical results

Design. 12 tasks x {baseline, APCI} x {non-reset, reset} = 48 runs, one run per cell, model labelled "Opus 7". Published as the ASE 2026 artifact; Zenodo DOI 10.5281/zenodo.19757261.

[FACT]

metric	baseline	APCI	change	per-task direction
non-reset code churn (mean)	576.58	955.42	+65.7%	APCI higher in 12/12
non-reset test churn (mean)	172.67	244.00	+41.3%	APCI higher in 12/12
reset true drift, delta CodeLOC (mean)	307.7	670.5	+117.9%	APCI lower in 1/12
CI pass	100%	100%	0.0 pp	48/48

[INTERPRETATION] Read at face value, v1 contradicts the hypothesis: APCI increased churn on every task and was more reset-inconsistent on 11 of 12. It should not be read at face value.

[LIMITATION] Seven recorded limitations, each of which independently undermines inference:

id	limitation	consequence
L1	the tree was never reset between tasks, so edits accumulate across T01-T12	runs are non-independent ; the churn trend is confounded with accumulation

id	limitation	consequence
L2	run_one_v1.sh does not invoke a model - it snapshots the working tree	generation and diff capture are not coupled
L3	AFCI-Guard matched literal libs/core paths while the code uses @afci-bench/* aliases	the guard's regexes never fired; its zeros are not measurements
L4	docs/ARCH_RULES.yml is 0 bytes	the machine-checkable architecture rules did not exist
L5	layer_jaccard is a constant 1.0	the metric cannot vary, so it measures nothing
L6	no per-task acceptance oracle beyond npm run ci, which is saturated True	the success gate does not discriminate
L7	npm ci fails on the clean v1 base	deterministic install is broken at the base tag

Also: the v1 baseline ran against a repository whose architecture was visible in the tree, so the "no architecture context" arm was not architecture-free.

v1 is historical / exploratory evidence. No v1 number may be pooled with, or contrasted against, a v2 number.

5. Architecture diagnostic results

These three experiments measure **instruments**, not treatments. A run-purpose firewall enforced by the record schema keeps them out of every confirmatory dataset, treatment-effect estimate and power calculation.

[FACT]

task	runs	functional	applicable opportunities	target violations	classification	consequence
PT08	3	3/3 (15/15 acceptance cases each)	3	0/3	functional ceiling + architecture floor	PT08 = REVISE; benchmark = CONTINUE
PT09	3	3/3 (14/14 each)	3	0/3	FAIL / ARCHITECTURE FLOOR	STOP / REASSESS
PT10	3	3/3 (8/8 each)	3	1/3	REVISE / WEAK PRESSURE	STOP / REASSESS

PT10's single violation is the only target architecture violation recorded anywhere in the programme: one forbidden import, on one line, in one already-existing file, severity blocker, confidence certain, automated. The opportunity id, rule id, forbidden scopes and anchor path are hidden evaluator semantics and stay in the private evaluator repository. **No numeric result is withheld.**

[INTERPRETATION] Three independent instruments at or near the architecture floor on this substrate is evidence about the **substrate**, not only about the tasks. A baseline at the floor has no room beneath it, so no treatment however good can be discriminated.

[LIMITATION] Three observations per instrument. No power calculation justifies that count and none is implied.

6. Sonnet efficiency pilot (Attempt 2)

Design. 3 tasks (PT01, PT04, PT07) x {C1, C4} x {non-reset, reset} x 3 repetitions = 36 runs, claude-sonnet-5, CLI 2.1.229, block-paired with randomised within-block condition order.

[FACT] Execution and validity:

	value
scheduled / executed	36 / 36
completed	35
refused	1 (sequence 12)
functionally valid	34 - C1 17/18 , C4 17/18
paired-eligible blocks	16 of 18 (minimum required 12)
runs entering the paired analysis	32
valid but unpaired	2

[FACT] Endpoints. Every figure is the median of per-block C4/C1 ratios; 1.0000 is parity and above 1.0000 means C4 cost more.

endpoint	n	median C4/C1	C4 lower	PT01	PT04	PT07
TOTAL_INPUT_TOKENS (primary)	16	1.4014	4/16	1.3372	0.9325	1.5419
MODEL_WALL_SECONDS	16	1.2660	4/16	1.2815	0.9834	1.7018
EXPLORATION_CALLS	16	1.3939	3/16	1.0714	1.3333	1.5130
TOTAL_TOOL_CALLS	16	1.1864	5/16	1.1000	0.9200	1.3867
UNIQUE_FILES_READ	16	1.2111	0/16	1.1429	1.3333	1.3571
EDIT_AND_WRITE_CALLS	16	1.9167	0/16	1.6000	1.3333	2.0000
CI_COMMAND_RUNS	16	1.0000	1/16	1.0000	1.0000	1.0000
TOTAL_OUTPUT_TOKENS (non-reset only)	9	1.4671	1/9	1.4671	1.0423	2.5088
provider cost USD (non-reset only)	9	1.4928	2/9	1.4438	0.9511	1.8851

By arm, on the primary endpoint: **NON_RESET 1.5582** (n=9) and **RESET 1.3776** (n=7). PT04 is the only task where C4 was cheaper on median.

[FACT] One signal ran the other way. Reset overhead is a run's metric under reset divided by the same task and condition without reset; lower is better.

endpoint	tasks where C4's recovery overhead is lower
TOTAL_INPUT_TOKENS	2 of 3 (PT01, PT07)
MODEL_WALL_SECONDS	3 of 3 (PT01, PT04, PT07)
EXPLORATION_CALLS	2 of 3 (PT01, PT07)
TOTAL_TOOL_CALLS	2 of 3 (PT01, PT07)

[FACT] The frozen decision. Both functional guardrails and the minimum-pairs gate passed; every token and secondary-endpoint threshold failed. The RESET-SPECIFIC GO branch failed only on its overall token clause - its two reset-overhead clauses passed.

Outcome (rule 11.4): STOP - NO EFFICIENCY SIGNAL JUSTIFIES FULL-SUITE EXPANSION. `efficiency_claim_made = false`.

[LIMITATION] AFCI_EFFICIENCY_PILOT is a **cost-only** purpose under its own frozen governance. **No architecture endpoint was produced and none may be inferred from it.** LOC churn was likewise not captured.

7. Haiku lower-model pilot

Design. The same three tasks on `claude-haiku-4-5-20251001`, non-reset only, 3 repetitions, 18 runs. The point was to test the most obvious explanation for the Sonnet null - that a strong model already knows what the MAD would tell it. Quality and efficiency are two **independent** channels, combined into no single score.

[FACT] Execution and validity: 18 of 18 executed, 18 COMPLETE, **0 refused, 0 infrastructure-invalid**. 17 of 18 functionally valid - C1 **8/9**, C4 **9/9**. The single invalid run, PT04/C1/R2, failed all four semantic acceptance cases and strands its valid C4 partner, leaving **8 of 9** paired-eligible blocks.

[FACT] Architecture quality:

arm	runs	applicable opportunities	violated	target-violation runs
C1	9	9	0	0
C4	9	9	0	0

Identical by task: 3 applicable, 0 violated in both arms for PT01, PT04 and PT07.

[FACT] Efficiency endpoints, 8 paired blocks:

endpoint	n	median C4/C1	C4 lower	PT01	PT04	PT07
TOTAL_INPUT_TOKENS (primary)	8	2.0372	2/8	2.9214	1.7502	0.9038
TOTAL_OUTPUT_TOKENS	8	1.7804	3/8	2.5067	1.4358	0.7171
MODEL_WALL_SECONDS	8	1.5268	3/8	2.3801	1.2307	0.8427
TOTAL_TOOL_CALLS	8	1.5023	2/8	2.0000	1.4174	0.9412
EXPLORATION_CALLS	8	1.7308	1/8	2.0000	2.1683	1.0769
UNIQUE_FILES_READ	8	1.2500	2/8	1.0000	1.0833	1.5000

endpoint	n	median C4/C1	C4 lower	PT01	PT04	PT07
EDIT_AND_WRITE_CALLS	8	1.1741	2/8	2.0000	1.1741	0.8750
CI_COMMAND_RUNS	8	1.0000	3/8	1.0000	0.7500	0.5000
provider cost USD	8	1.8865	2/8	2.5232	1.5777	0.8078

PT07 is the one task where C4 was directionally cheaper on tokens, time, output and cost. PT01 was the most expensive for C4 on every endpoint.

[FACT] Rework over the 8 paired blocks: edit calls 54 vs 66, files re-edited 16 vs 20, turns used 229 vs 309, files changed in the worktree diff 25 vs 30, net lines +1,621 vs +1,746. Agent-initiated CI command runs went the other way: 13 for C1 against 8 for C4.

[FACT] The frozen decision:

signal	criterion	observed	verdict
QUALITY	12.1.1 C4 has fewer target architecture-violation runs overall	C4 0 vs C1 0	FAIL
QUALITY	12.1.2 improvement in at least 2 of 3 tasks	improved: none	FAIL
QUALITY	12.1.3 functional guardrail	C4 valid 9 vs C1 valid 8; deficit -1 <= 1	PASS
EFFICIENCY	12.2.1 functional guardrail	C4 valid 9 vs C1 valid 8; deficit -1 <= 1	PASS
EFFICIENCY	12.2.2 at least one endpoint threshold met	median TOTAL_INPUT_TOKENS 2.0372 < 1.00; median EXPLORATION_CALLS 1.7308 <= 0.80; median TOTAL_TOOL_CALLS 1.5023 <= 0.85	FAIL

NO LOWER-MODEL SIGNAL - DO NOT EXPAND THE SYNTHETIC LOWER-MODEL MATRIX

[LIMITATION] Criterion 12.1.1 fails on a **tie at zero**, not on C4 being worse. Neither arm violated anything, so there was nothing to discriminate. The rule requires *strictly fewer* violations, so a tie scores FAIL. That is a rule outcome, not a finding about AFCI: the quality channel produced **no information**.

8. Cost and token findings

[FACT] Token audit. TOTAL_INPUT_TOKENS = input_tokens + cache_creation_input_tokens + cache_read_input_tokens on every row that carries tokens - checked on all 53 such rows, 0 failures. The MAD's own tokens and all cache traffic stay inside the number.

scope	paired blocks	C1 total input tokens	C4 total input tokens	C1 median/run	C4 median/run	median C4/C1
Sonnet overall	16	10,988,803	14,859,860	698,639	1,040,459	1.4014
Sonnet NON_RESET	9	5,912,725	8,339,159	664,777	1,062,096	1.5582
Sonnet RESET	7	5,076,078	6,520,701	733,633	1,028,101	1.3776
Haiku NON_RESET	8	5,599,750	9,504,643	652,919	1,271,816	2.0372

Token coverage: 53 of 121 run rows carry input-token evidence, 37 carry output tokens. v1, the aborted Attempt 1 and the three diagnostics predate or do not use the token instrumentation; their cells are blank, never 0.

[FACT] Cost audit.

scope	runs	runs with complete cost	captured total	verdict
Sonnet Attempt 2 - all rows	36	19	\$11.1919	PARTIAL COST CAPTURE
Sonnet Attempt 2 - NON_RESET	18	18	\$10.9463	COMPLETE COST COVERAGE
Sonnet Attempt 2 - RESET	17	1	\$0.2456	PARTIAL COST CAPTURE
Haiku lower-model pilot - all rows (NON_RESET by design)	18	18	\$3.8007	COMPLETE COST COVERAGE

The Sonnet RESET row counts 17, not 18: the refused run (sequence 12) carries no reset state at all, so it sits outside both arms. 18 + 17 + 1 = 36.

paired cost ratio	n	C1 total	C4 total	median C4/C1
Sonnet NON_RESET	9	\$4.5288	\$6.4175	1.4928
Sonnet RESET	0	n/a	n/a	no cost comparison exists
Haiku NON_RESET	8	\$1.3862	\$2.0068	1.8865

Total captured provider cost across all runs carrying cost evidence: \$14.99 over 37 runs.

CAPTURED COST, NOT NECESSARILY TOTAL STUDY COST. 84 of 121 run rows carry no provider-cost record at all: all 48 v1 rows, all 9 Attempt-1 rows, all 10 diagnostic rows, 16 of 17 Sonnet RESET rows, and the one refused run. Money was spent on those runs; the runtime never reported it in a form the record could carry.

[LIMITATION] A reset run's phase A is deliberately interrupted at the checkpoint, **before** its terminal result event - the only place the runtime reports output tokens and cost. For a reset run, input tokens remain **exact** (reconstructed from the streamed assistant messages) while output tokens and cost are **withheld, not zero and not estimated**. The one Sonnet reset run that does carry a cost is PT04/C1/RESET/R1, and it carries one precisely because it never reached its checkpoint - it is functionally invalid and enters no paired figure. There are therefore **zero usable Sonnet RESET cost pairs**, and the 1.4928 figure describes the non-reset arm only.

9. Functional and architecture quality

[FACT]

experiment	C1 valid	C4 valid	paired-valid blocks	semantic cases passed (C1 / C4)
Sonnet Attempt 2	17/18	17/18	16 of 18	68 / 68
Haiku lower-model pilot	8/9	9/9	8 of 9	32 / 36

[FACT] Architecture, over every run that produced a measurement:

experiment	arm	runs	applicable	violated	target-violation runs
PT08 diagnostic	C1	3	3	0	0
PT09 qualification	C1	3	3	0	0
PT10 qualification	C1	3	3	1	1
Haiku pilot	C1	9	9	0	0
Haiku pilot	C4	9	9	0	0
Sonnet Attempt 2	both	36	NOT MEASURED	NOT MEASURED	NOT MEASURED
V1 original	both	48	INVALID FOR INFERENCE	INVALID FOR INFERENCE	INVALID FOR INFERENCE

[INTERPRETATION] Functional acceptance is saturated and architecture sits at a floor. Six instrument/model combinations now sit at or near that floor on this substrate: PT08, PT09 and PT10 for an unguided Sonnet baseline, and PT01, PT04 and PT07 for Haiku in **both** arms.

[LIMITATION] The evidence base is lopsided. The study's *actual* construct - architectural conformance - has been measured only by instruments that all sat at or near the floor, while its *secondary* construct - cost - has one clean pilot. The cost result must not be allowed to stand in for an architecture result.

10. Cross-model interpretation

[FACT] Both sides below are NON_RESET, so the comparison is like-for-like.

metric	Sonnet C4/C1 (n=9)	Haiku C4/C1 (n=8)	difference	reading
TOTAL_INPUT_TOKENS	1.5582	2.0372	+0.4791	Haiku ratio higher
TOTAL_OUTPUT_TOKENS	1.4671	1.7804	+0.3133	Haiku ratio higher
MODEL_WALL_SECONDS	1.4968	1.5268	+0.0300	Haiku ratio higher
EXPLORATION_CALLS	1.5714	1.7308	+0.1593	Haiku ratio higher
TOTAL_TOOL_CALLS	1.3333	1.5023	+0.1690	Haiku ratio higher
UNIQUE_FILES_READ	1.4000	1.2500	-0.1500	Haiku ratio lower
EDIT_AND_WRITE_CALLS	2.0000	1.1741	-0.8259	Haiku ratio lower
PROVIDER_COST_USD	1.4928	1.8865	+0.3937	Haiku ratio higher

channel	Sonnet	Haiku	comparable?
architecture target violations	NOT MEASURED (cost-only purpose)	0/9 in C1 and 0/9 in C4	no
functional validity	17/18 and 17/18	8/9 and 9/9	descriptively only

This is a descriptive comparison between two separate experiments, not a randomised cross-model causal comparison. The two pilots ran under separate purposes, separate schedules and separate frozen analyses and are never pooled. No interaction was estimated, no test was performed, and no pooled model exists.

[INTERPRETATION] The moderator hypothesis predicted that a weaker coding model, having less ability to infer architecture from the repository alone, would find the explicit MAD *more* useful. The prediction did not come true on the primary endpoint: C4's median input-token ratio **rose** from 1.5582 to 2.0372, a difference of 0.4791 in the direction opposite to the prediction, and the quality channel produced nothing at all because both arms sat at zero.

Two of the eight metrics did move the predicted way - `UNIQUE_FILES_READ`, `EDIT_AND_WRITE_CALLS` - and both are reported above rather than dropped. Neither is the primary endpoint, both remain **above** 1.0000 on the Haiku side (so C4 still cost more in absolute terms on both models), and with 8 and 9 paired blocks and no test performed, a two-of-eight split is not a result. It is recorded so that the six-of-eight direction is not read as unanimity.

11. What the evidence supports

1. **[FACT]** On this substrate with `claude-sonnet-5`, explicit MAD injection costs more than it saves across every captured cost dimension except CI invocations.
2. **[FACT]** It does so without buying functional quality: 17/18 versus 17/18.
3. **[FACT]** Weakening the model did not reduce C4's relative cost; it raised it, and produced no quality gain because both arms sat at zero violations.
4. **[FACT]** The reset-recovery direction is the one consistent counter-signal, strongest in wall time (3 of 3 tasks).
5. **[INTERPRETATION]** The **strong-model ceiling** explanation for the repeated null is **not supported**. Instrument discrimination, not model capability, is the binding constraint on the programme.
6. **[INTERPRETATION]** Expanding either pilot, unchanged, is not justified. Both frozen rules returned a negative verdict and both were honoured.
7. **[INTERPRETATION]** The benchmark machinery works. Sterile execution, model pinning and readback, context auditing, hidden acceptance, the architecture oracle, the artifact firewall, run-identity derivation and the frozen analysis all executed end to end - and caught their own defects twice, in ways that cost real money and were recorded rather than smoothed over.

12. What the evidence does NOT support

1. **We cannot conclude AFCI does not work.** We can conclude it did not reduce cost here. Efficiency is one construct; architectural conformance is the study's actual construct, and the cost pilot **did not measure it at all**.
2. **We cannot attach any statistical confidence to anything.** No p-value, no confidence interval, no effect size, no power estimate.
3. **We cannot generalise past this substrate** - one synthetic 49-file monorepo whose architecture is legible from its own code.
4. **We cannot generalise past the two models used.** `primary_model` is still null; no model has been selected for the study.
5. **We cannot use the reset-recovery signal as a finding.** It is descriptive, from 3 tasks, and it failed the branch that would have made it a GO.
6. **We cannot read anything as a v1 hypothesis test.** L1-L7 make the v1 comparison uninterpretable as a treatment contrast.
7. **We cannot claim any architecture result from the Sonnet pilot.** It produced none.
8. **We cannot claim an architecture result from the Haiku pilot either.** It produced an endpoint, but at the floor in both arms. A tie at zero discriminates nothing and must not be read as AFCI failing to improve architecture.
9. **We cannot claim that model capability moderates AFCI's value.** Two separate experiments compared descriptively, never pooled.

13. Limitations

#	limitation
1	No confirmatory evidence exists. Collection has not begun. The suite-wide protocol is PRE-FREEZE, gate G1 is not passed, and TD - B32, TD - B34, TD - B03 and TD - B19 remain open.
2	No statistical inference of any kind. Every reported figure is a median of paired ratios - a descriptive statistic. No power calculation has ever been run.
3	One synthetic substrate , 49 files, whose layering is inferable from its own import graph and path aliases. A model can often deduce the intended architecture <i>without</i> the MAD, which directly attacks the MAD's marginal value. This is the primary external-validity threat to every v2 finding.
4	No study model has been selected. Both models used so far are provisional.
5	Architecture is measured only where it sat at a floor. The evidence base is lopsided toward cost.
6	Reset cost and output tokens do not exist for reset runs - withheld by construction, not missing by accident.
7	This package is a secondary artifact. Where it and a primary artifact disagree, the primary artifact wins.

14. Current research direction

The chain is short and each link is recorded:

1. **v1 measured the wrong things well enough to notice.** Churn is a proxy, the guard did not fire, the oracle did not discriminate, and runs were not independent. v2 replaced proxies with a hidden functional acceptance oracle and an out-of-band architecture oracle.
2. **PT08 showed the instruments, not the theory, were the bottleneck.** A baseline at the architecture floor cannot discriminate however good the treatment is.
3. **PT09 and PT10 showed it was not specific to PT08.** Two purpose-built replacements landed in the same place.
4. **The efficiency pilot asked a cheaper question that could be answered now** - does the MAD at least pay for itself? - and answered **no**, under a rule frozen before the data existed.
5. **The lower-model pilot weakened the model, and the null held.** A weaker model produced no quality signal and a *worse* cost ratio.
6. **[INTERPRETATION]** The common factor across steps 2-5 is not model strength - that has now been varied and did not matter. What remains is that **a legible synthetic repository gives an agent little to gain from being told its architecture**, whether the agent is strong or weak.

[INTERPRETATION] Repository architectural complexity / ambiguity / context-recovery burden is therefore the next scientifically distinct moderator to investigate. **This is the next hypothesis to test, not a proven one.** It stands by elimination rather than by evidence, and nothing in this package tests it.

15. Next study - open-source architectural complexity

Status: NOT STARTED. No runs exist. No results are pre-populated.

Re-run against real / open-source repositories with genuine architectural complexity, where the architecture is *not* inferable from a 49-file import graph. That directly tests the substrate-legibility explanation - the only one of the two rival explanations still standing.

Its motivation was never contingent on the lower-model outcome: repository complexity is an independent moderator and the lower-model pilot measured nothing about it. Removing the ceiling rival raises its priority; it does not authorise it.

Before any run it needs its own decision record, new tasks, new hidden acceptance packages and new architecture rule linkage. Gate G1 is still not passed and the suite is still not frozen.

Sources: study-results/AFCI_MASTER_RUN_RESULTS.csv, AFCI_MASTER_EXPERIMENT_REGISTRY.csv, and the frozen per-experiment analysis artifacts under 01_v1_original_study/ through 06_lower_model_pilot_completed/. Traceability from any figure here to its raw artifact is in AFCI_Professor_Evidence_Map.md and workbook sheet 18_EVIDENCE_MAP.